20. $\bar{x}_1 = 5$ $s_1^2 = 0.025$ $n_1 = 5$ C105503Q20

$\bar{x}_2 = 4.86$ $s_2^2 = 0.013$ $n_2 = 5$

a. $H_0: \mu_1 = \mu_2$

$H_a: \mu_1 > \mu_2 \Rightarrow \mu_1 - \mu_2 > 0$

$\alpha = 0.05$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = 0.019$$

$$\Rightarrow s_p = \sqrt{0.019} = 0.1378$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{(\frac{1}{n_1} + \frac{1}{n_2}) s_p^2}} = \frac{0.14}{\sqrt{1.019(\frac{1}{5}+\frac{1}{5})}} = 1.61$$

$t_{8, 0.05} = 1.86$

$1.61 < 1.86 \Rightarrow$ Non-reject H_0

b. $\mu_1 - \mu_2$ 的 $100 \times (1 - 0.05)\%$ C.I.

$t_{8, 0.025} = 2.306$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{n_1+n_2-2, \frac{\alpha}{2}} \sqrt{(\frac{1}{n_1} + \frac{1}{n_2}) s_p^2}$$

$$= 0.14 \pm 2.306 \sqrt{\frac{2}{5} \times 0.019} \Rightarrow (-0.061, 0.341)$$

包含 0 ⇒ 上下游平均溶氧量可能相同

C10504Q09.

$d = x_1 - x_2$

$d_1 = 0.1$
$d_2 = 0.1$
$d_3 = 0$
$d_4 = 0.2$
$d_5 = -0.1$

$\bar{d} = 0.06$
$S_d = 0.114$

① $H_0: \mu_1 - \mu_2 = \mu_d = 0$
$H_a: \mu_d \neq 0$
$\alpha = 0.05$

$$T = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{0.06}{0.114/\sqrt{5}} = 1.177$$

$t_{4, 0.025} = 2.776$

$1.177 \not> 2.776$, Non-reject H_0

② 95% C.I.

$$\bar{d} \pm t_{n-1, \frac{\alpha}{2}} \cdot \frac{S_d}{\sqrt{n}}$$

$$= 0.06 \pm 2.776 \times \frac{0.114}{\sqrt{5}} \Rightarrow (-0.082, 0.202)$$

③ 包含0 ⇒ $\mu_1 - \mu_2$ 可能 = 0
Non-reject H_0

C10504Q20

$d_1 = 15$
$d_2 = 15$
$d_3 = 15$
$d_4 = 9$
$d_5 = 16$
$d_6 = 8$
$d_7 = 12$
$d_8 = 8$
$d_9 = 10$
$d_{10} = 12$
$d_{11} = 9$
$d_{12} = 4$
$d_{13} = 10$
$d_{14} = 4$
$d_{15} = 17$
$d_{16} = 13$
$d_{17} = 4$
$d_{18} = 7$
$d_{19} = 7$
$d_{20} = 10$

$\bar{d} = 10.25$
$S_d = 4.05$

$H_0: \mu_1 - \mu_2 = \mu_d = 0$
$H_a: \mu_d > 0$
$\alpha = 0.05$

$$t = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{10.25}{4.05/\sqrt{20}} = 11.318$$

$t_{19, 0.05} = 1.729$

$11.318 \gg 1.729 \Rightarrow$ Reject H_0

→ 平均 recall is higher

C10505Q13

a. $H_0: \sigma = 0.7$
$H_a: \sigma > 0.7$

$\bar{x} = 352.5$
$S = 1.915$
$S^2 = 3.667$

$$\chi^2 = \frac{3 \times 3.667}{0.7^2} = 22.451$$

$\chi^2_{3, 0.05} = 7.814$

$22.451 > 7.814 \Rightarrow$ Reject H_0

b. C.I. 90%

$$\left[\frac{3 \times 3.667}{\chi^2_{3, 0.05}}, \frac{3 \times 3.667}{\chi^2_{3, 0.95}}\right]$$

$$= (1.41, 31.25)$$

C10S04Q22

$d_1 = 0.006$
$d_2 = -0.018$
$d_3 = -0.014$
$d_4 = -0.011$
$d_5 = -0.004$
$d_6 = -0.019$

$\bar{d} = -0.01$
$S_d = 0.00953$

a. $H_0: \mu_1 - \mu_2 = \mu_d = 0$
$H_a: \mu_d \neq 0$
$\alpha = 0.05$

$$t = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{-0.01}{0.00953/\sqrt{6}} = -2.5703$$

$t_{5, 0.025} = 2.571$

$|t| = 2.5703 < 2.571$, Non-reject H_0

b.
$$-0.01 \pm 2.571 \times \frac{0.00953}{\sqrt{6}}$$
$$\Rightarrow (-0.020, -0.000)$$

C10S05Q12

a. Completions $\bar{x} = 22.5$
$s = 4.705$

Yards $\bar{x} = 12.9875$
$s = 2.531$

b. $df = 15$

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1,\frac{\alpha}{2}}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}\right] = \left(\frac{332}{27.488}, \frac{332}{6.262}\right)$$
$$= (12.078, 53.018)$$

c. 95% C.I. for σ^2

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1,\frac{\alpha}{2}}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}\right]$$
$$= \left[\frac{15 \times 2.531^2}{27.488}, \frac{15 \times 2.531^2}{6.262}\right]$$
$$= (3.496, 15.345)$$

for σ

$$(\sqrt{3.496}, \sqrt{15.345})$$
$$= (1.870, 3.917)$$

d. $H_0: \sigma = 4$,
$H_a: \sigma \neq 4$

$$\chi^2 = \frac{15 \times 2.531^2}{4^2} = 6.005$$

$\chi^2_{0.025} = 27.488$ $\chi^2_{0.975} = 6.262$

$6.005 < 6.262 \Rightarrow$ Reject H_0